MTH 201 **Skills Check 1 Worksheet**

Skills checks cover pre-requisite material needed for the first unit of the course, areas & volumes. Answer the following questions neatly and completely. Calculators are not allowed.

1. Shade the region defined by the following curves: $f(x)=\ln(x)$, $y=0$, $x=e$.

2. Shade the region bounded by the following curves: $x=y^2-1$ & $x=3$.

3. Shade the region bounded by the following curves: $f(x)=2x-1$, $x=5$, $x=3$, & $y=0$.

4. Find the points of intersection of the curves: $x = 2y + 3$ & $x = y^2$.

5. Find the points of intersection of the curves: $y = x^3$ & $y = \sqrt{x}$.

6. Find the points of intersection of the curves: $y = x^2$, $y = 4 - x^2$.

7. Evaluate the definite integral: $\int_{-3}^{-2}(x+3)^2\,dx$.

8. Evaluate the definite integral: $\int_{2}^{5}\pi\left(\frac{1}{x}\right)^2 dx$.

9. Evaluate the definite integral: $\int_{1}^{2}\pi\left(k+\frac{1}{k}\right)dk$.

10. Use a u-substitution to evaluate the definite integral: $\int_{-2}^{1} \frac{x}{2x^2+1}\,dx$.

11. Use a u-substitution to evaluate the definite integral: $\int_{0}^{\pi/3} 3\sin(x)e^{\cos(x)}\,dx$.

12. Use a u-substitution to evaluate the definite integral: $\int_{0}^{1} \pi t\left(t^2-1\right)^5\,dt$.

13. Solve the equation for x: $y = \sqrt[3]{2x+1}$.

14. Solve the equation for x: $y = 1 - \ln(x)$.

15. Solve the equation for x: $y = \dfrac{1}{2-x}$.